

COMPUTER NETWORKS

Bandwidth, Delay, Throughput & Latency

Numerical Problems

LEARNING OBJECTIVES

- • Understand Bandwidth
- • Calculate Propagation Delay
- • Compute Throughput
- • Find Network Latency

BANDWIDTH

- Bandwidth = Maximum data transfer rate
- Unit: bps (bits per second)
- Example: 10 Mbps

BANDWIDTH FORMULA

- $\text{Data} = \text{Bandwidth} \times \text{Time}$
- $\text{Time} = \text{Data} / \text{Bandwidth}$

Problem 1

A network link has a bandwidth of **20 Mbps**.
How much data (in MB) can be transmitted in **10 seconds**?

Solution:

$$\begin{aligned}\text{Data} &= \text{Bandwidth} \times \text{Time} \\ &= 20 \text{ Mbps} \times 10 \text{ s} \\ &= 200 \text{ Mb} = \mathbf{25 \text{ MB}}\end{aligned}$$

Problem 2

A file of size **500 MB** is transmitted over a **50 Mbps** link.
Find the minimum transmission time.

Solution:

$$500 \text{ MB} = 4000 \text{ Mb}$$

$$\begin{aligned} \text{Time} &= \text{Data} / \text{Bandwidth} \\ &= 4000 / 50 = \mathbf{80 \text{ seconds}} \end{aligned}$$

PROPAGATION DELAY

- Time taken by signal to travel
- Depends on distance and speed
- $\text{Delay} = \text{Distance} / \text{Speed}$

Problem 3

Distance between two nodes is **1000 km**.

Signal speed = **2×10^8 m/s**.

Find the propagation delay.

Solution:

Distance = 1000 km = 10^6 m

Delay = $10^6 / (2 \times 10^8)$

= **0.005 s = 5 ms**

Problem 4

A signal travels through an optical fiber of length **200 km** at **2.5×10^8 m/s**.

Calculate propagation delay.

Solution:

Delay = $2 \times 10^5 / (2.5 \times 10^8)$

= **0.0008 s = 0.8 ms**

THROUGHPUT

- Actual data transfer rate
- Usually less than bandwidth
- $\text{Throughput} = \text{Data} / \text{Time}$

Problem 5

A file of **100 MB** is downloaded in **40 seconds**.
Find the throughput.

Solution:

$$100 \text{ MB} = 800 \text{ Mb}$$

$$\begin{aligned}\text{Throughput} &= 800 / 40 \\ &= \mathbf{20 \text{ Mbps}}\end{aligned}$$

Problem 6

A network has bandwidth of **100 Mbps**, but only **60 Mbps** is achieved.
Find the efficiency.

Solution:

$$\begin{aligned}\text{Efficiency} &= (60 / 100) \times 100 \\ &= \mathbf{60\%}\end{aligned}$$

LATENCY

- Total time taken by packet
- Includes multiple delays
- $\text{Latency} = \text{Transmission} + \text{Propagation} + \text{Processing} + \text{Queuing}$

Problem 7

A packet of size **1 MB** is sent over a **10 Mbps** link. Distance = **500 km**

Speed = **2×10^8 m/s**

Processing = **2 ms**

Queuing = **3 ms**

Find total latency.

Solution:

1 Transmission Delay

1 MB = 8 Mb

$= 8 / 10 = \mathbf{0.8\ s}$

2 Propagation Delay

500 km = 5×10^5 m

$= 5 \times 10^5 / 2 \times 10^8$

$= 0.0025\ \text{s} = \mathbf{2.5\ ms}$

3 Total Latency

$= 0.8 + 0.0025 + 0.002 + 0.003$

$= \mathbf{0.8075\ s}$

Problem 8

A packet of **500 KB** is transmitted over **5 Mbps** link.

Propagation delay = **4 ms**

Processing delay = **1 ms**

Queuing delay = **2 ms**

Find total delay.

Solution:

500 KB = 4 Mb

Transmission = $4 / 5 = \mathbf{0.8\ s}$

Total = $0.8 + 0.004 + 0.001 + 0.002$
 $= \mathbf{0.807\ s}$

PRACTICE PROBLEM

Problem 9

A 200 MB file is sent over a 40 Mbps link. Find transmission time.

Problem 10

Distance = 800 km; Speed = 2×10^8 m/s; Find propagation delay.

Problem 11

A user downloads 300 MB in 50 seconds. Find throughput.

Problem 12

Packet size = 2 MB; Bandwidth = 20 Mbps; Distance = 1000 km; Speed = 2×10^8 m/s
Processing = 1 ms; Queuing = 2 ms; Find latency.

SUMMARY

- • Bandwidth = Capacity
- • Delay = Time
- • Throughput = Performance
- • Latency = Total Delay